Inventory Stratification with Safety Stock & ROP Determination

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Introduction

Efficient inventory management helps achieve higher

PROFITABILITY

How?

Minimizes logistics costs	
Reduces risks	
• **Supply side risk**	• Demand side risk

Data Overview

Inventory Classification

Given: List of digital consumer electronics items

- Sales
- Hits

- Sales for each item individually ranked and scored
- Hits for each item individually ranked and scored
- Scores combined and weighted for each item
- Combined final rank calculated for each item

Safety Stock and ROP Determination

- Inventory classification complete
- Service level determined for digital consumer electronics
- Safety stock calculated
- Reorder point calculated

Inventory Classification Methodology

1

Calculate cumulative and % sales

2

Rank items individually A,B, C, or D; A, most important, and D, least important

3

Calculate weighting and spread

4

Evaluate data for a final ranking

Inventory Classification Methodology

Ranking Strategy	
Rank	Level
A	60%
B	80%
C	90%
D	100%

Weighting Strategy	
Sales	65%
Hits	35%

Spread		
Rank	From	To
A	3.25	4.00
B	2.5	3.24
C	1.75	2.49
D	0	1.75

Sample Table: Sales Ranking

Item Number	Sales ($)	Cumulative Sales	% of Sales	Rank
1090	9328940	9328940	11	A
1180	7202900	16531840	20	A
1196	1302104	49247848	60	B
1205	1161050	50408898	61	B
1070	400200	65767909	80	C
1209	398620	66166529	81	C
1150	303303	73994108	90	D
1175	297000	74291108	91	D

Sample Table: Hits Ranking

Item Number	Hits	Cumulative Hits	% of Hits	Rank
1092	2064	2064	9	A
1090	1780	3844	17	A
1166	345	14251	61	B
1164	342	14593	63	B
1199	86	18631	80	C
1195	86	18717	80	C
1053	71	20990	90	D
1153	68	21058	91	D

Sample Table: Final Ranking

Item Number	Sales Rank	Hits Rank	Sales Rank Score	Hits Rank Score	Weighted Score	Final Rank
1090	A	A	4	4	4.0	A
1170	A	A	4	4	4.0	A
1215	B	B	3	3	3.0	B
1036	B	C	3	2	2.7	B
1209	C	C	2	2	2.0	C
1160	C	B	2	3	2.4	C
1150	D	B	1	3	1.7	D
1024	D	D	1	1	1.0	D

Safety Stock & Reorder Point (ROP)

Digital consumer electronics retailer must run higher service levels

- These are high value items
- Demand is often variable and influenced by promotions or seasonality
- Stockouts result in lost sales, not production, and customers switch instantly if item is out of stock

Service level strategy:

Rank	Service Level
A	98%
B	95%
C	88%
D	70%

Sample Table: Safety Stock & Reorder Point (ROP)

Item Number	Final Rank	Service Level	Standard Normal Variate	Safety Stock	ROP
1170	A	0.98	2.05	62.50	**125.50**
1036	B	0.95	1.64	61.13	**113.13**
1160	C	0.88	1.17	73.82	**157.82**
1024	D	0.70	0.52	18.29	**48.29**

Summary & Conclusion

Inventory stratification was performed for all given items

A service level was chosen and safety stock levels and the reorder points for the four specified items was calculated.

Counts of final rank and sum of sales per rank are reported

Final Rank	Count of Final Rank
A	22
B	18
C	23
D	142
Grand Total: 205	

Final Rank	Sum of Sales ($)
A	54,788,937
B	10,450,897
C	8,173,036
D	8,633,280
Grand Total: 82,046,150	